

Nishal Thomas

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Education

St. Mary's Sr. Sec. School

Class 12

Class 10 [CBSE] — 91%

New Delhi, India

Ongoing

2025

Research

FormInv: A Measurement Protocol for Semantic Invariance in Mathematical Reasoning Benchmarks [\[arXiv\]](#)

[First Author • ICML 2026 AI4Math Workshop (Poster)]

- First author of an accepted ICML 2026 workshop paper introducing FormInv, a benchmark and evaluation framework for measuring semantic invariance in mathematical reasoning systems
- Developed a dataset of Lean4-grounded theorem paraphrases spanning 8 transformation families and evaluated robustness across 9 frontier language models
- Proposed Semantic Consistency Rate (SCR) and cross-model unanimity auditing, revealing formulation-sensitive failure modes and benchmark construction errors hidden by aggregate accuracy metrics

When Models Know Without Representing: Behavioural–Representational Dissociation in LLM Abstraction Hierarchies [\[Workshop Submission Under Review\]](#) [\[Code\]](#)

- Constructed 50 abstraction ladders spanning nine levels of conceptual abstraction across mathematics and scientific domains
- Investigated behavioural–representational dissociation in language models using representation geometry, abstraction coherence metrics, and activation patching
- Conducted experiments across model scales from 70M to 70B parameters to study abstraction formation and scaling behaviour

Mapping Hallucinations in Tiny Aya

Multilingual Hallucination Detection Study

- Participated in Expedition Tiny Aya, a global research initiative by Cohere Labs, collaborating within a five-member international team to investigate multilingual hallucinations in Tiny Aya models
- Applied black-box metrics (Hallucination Rate, Self-Consistency, Variance, Cross-Model Disagreement, Prompt Sensitivity Score, etc.) alongside white-box methods to detect hallucination signals using Python-based pipelines

Technical Projects

Step-Free Arithmetic Transformers *Mechanistic Interpretability Study*

- Built a reproducible research framework for training compact arithmetic transformers (0.66M–10M parameters) on hierarchical mathematical reasoning tasks using PyTorch and mixed-precision training.
- Applied mechanistic interpretability techniques including attention-head ablation, linear probing, and out-of-distribution evaluation to identify parenthesis-sensitive computation circuits underlying arithmetic reasoning.

Aletheia *Evidence-Grounded LLM Assistant*

- Engineered a transparent retrieval-augmented reasoning pipeline using techniques from **Loki** (claim decomposition), **LM-Polygraph** (verbalized confidence estimation), **FIRE** (adaptive retrieval), and **OpenFactCheck** (evidence-grounded verification), integrating them into a unified FastAPI, Qdrant, Ollama, and Next.js system.
- Developed a self-enriching RAG architecture that automatically retrieves and indexes arXiv and Wikipedia content when retrieval coverage is insufficient, enabling continual expansion of the local knowledge base.

Grokking Feasibility Boundaries in Transformers *Generalization Dynamics Study*

- Constructed controlled interpolation–extrapolation benchmarks for modular arithmetic to isolate the effect of train–test overlap on grokking behaviour in compact transformer models.
- Investigated delayed generalization through representation geometry, representation rank collapse, and attention entropy, identifying empirical boundaries for grokking under varying supervision regimes.

Continuum [K2-Think Hackathon] *AI Safety Infrastructure*

- Designed an AI safety system for detecting epistemic drift and contradiction accumulation across long-running LLM conversations through cumulative stance tracking and dependency analysis.
- Implemented a hybrid verification pipeline combining symbolic heuristics with external reasoning models to identify, validate, and visualize conversational inconsistencies.

Extracurriculars

- Participated in multiple speaking competitions including speech and debate, securing awards and recognitions at inter-school level.
- Served as a member of the School Student Council, contributing to student representation and initiatives
- Qualified Stage 1 of the Indian National Olympiad of Artificial Intelligence (INAIO)

Awards

2025: YMCA Academic Excellence Award — Class 10 Board performance

2025: Science Topper — Class 10, St. Mary's Sr. Sec. School

2025: 100/100 in Artificial Intelligence — CBSE Board Examination